

Power wheelchair driving: a multisensory simulator using VR to learn in rehabilitation centers

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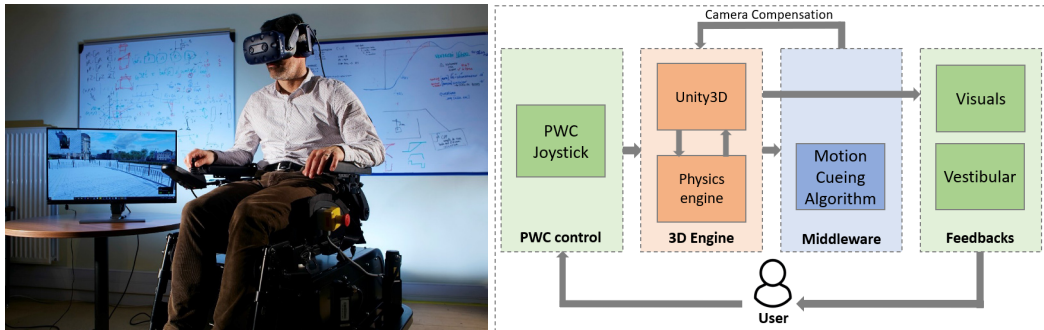


Figure 1: *left*: The simulator with a HMD. *Right*: The framework of the PWC simulation.

ABSTRACT

Power wheelchairs (PWCs) significantly enhance mobility for individuals with disabilities but are often challenging to master, requiring extensive training. Traditional training methods can be risky, resource-intensive, and difficult to implement. Virtual reality (VR) offers a safer, customizable, and effective alternative, as demonstrated in rehabilitation contexts. We developed a multisensory VR simulator incorporating vestibular feedback to enhance the sense of presence and mitigate cybersickness. Our studies show effective skill transfer from the virtual environment to real-world PWC use, validated with actual users in clinical trials. This demonstration highlights the capabilities of our mechanical simulator, featuring navigation scenes from previous trials and ongoing work with virtual agents. The simulator's immersive and adaptive design addresses the challenges of PWC training, offering a practical and innovative solution for clinicians and users alike.

Index Terms: Power wheelchair, Simulator, Multisensory Feedback, Clinical Environment

1 INTRODUCTION

Power wheelchairs (PWCs) are a widely adopted solution for individuals with disabilities to maintain autonomy in their everyday movements. These devices significantly enhance mobility and independence, but learning how to drive a PWC is a complex and demanding process. It requires not only motor coordination but also cognitive effort, as users must learn to navigate varied and often dynamic environments safely. Traditional PWC teaching methods are filled with challenges. These approaches can expose users to dangerous situations, such as collisions or tipping, especially during early learning stages. Additionally, setting up training sessions in real-world environments can be logistically difficult and resource-intensive. Such barriers can discourage effective learning and slow progress in gaining confidence and proficiency with PWCs.

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Virtual reality (VR) offers a promising alternative for training. [4] shows the efficiency of VR rehabilitation programs compared to traditional rehabilitation programs. By creating immersive and interactive environments, VR enables users to practice and refine their skills in a safe and controlled setting. The technology allows for customizable scenarios that can replicate real-world challenges without the associated risks, providing a versatile platform for skill acquisition.

In this demonstration, we present our multisensory PWC simulator which use VR to help therapists teaching PWC driving to end users in rehabilitation centers.

2 A MULTISENSORY POWER WHEELCHAIR SIMULATOR

Our multisensory PWC simulator [9] was co-designed with clinicians and PWC users based on end-users needs. Therefore, the final system respect several constraints:

- It has a similar form factor and seating to a real PWC,
- It can be used with user-specific controllers available on the market,
- It reproduces accurate motions simulating the sensations of a real power wheelchair though the motion cueing algorithm,
- It is compliant with any XR device,
- It minimizes the cybersickness, coming from perception conflicts between visual, vestibular and proprioceptive systems.

The simulation relies on the framework shown on Fig. 1. For the demonstration, the user interface is a standard RNET joystick commonly used in commercial PWC, which is used in a PWC controller to drive a simulated PWC in a virtual environment in Unity3D. The simulator's motion cueing algorithm controls the mechanical platform based on the movements of the simulated PWC. This vestibular and haptic feedback, associated with a proper HMD motion compensation effectively minimizes cybersickness and ensures a more immersive and comfortable experience. A more detailed description of the framework is given in our previous work [9].

3 RESULTS

In our initial work [2], we demonstrated the benefits of VR for PWC training, showing that even simple VR simulations can provide a strong Sense of Presence. However, a common challenge in VR is cybersickness, which negatively impacts user experience, training quality, and Sense of Presence [2, 1, 5]. To address this, we collaborated with clinicians to design a multisensory VR simulator specifically tailored for PWC training [9]. This simulator integrates vestibular and haptic feedback alongside visual stimuli, which have been shown to significantly enhance the sense of presence in virtual environments [8].

Before exploring social navigation in VR, it was necessary to validate that driving performance in VR closely matches real-life conditions and that cybersickness remains tolerable. A 2020 study [3] involving 32 participants with neurological impairments revealed similar performance on simple circuits but noted more collisions and longer completion times in VR for complex circuits, alongside a higher mental workload. In 2021, a follow-up study compared four interfaces (two immersive and two non-immersive) with 25 experienced PWC users. Immersive systems, such as CAVE and VR headsets, demonstrated better performance and increased presence but induced higher levels of cybersickness.

Additionally, a 2024 study [6] aimed to model locomotor behaviors, particularly interpersonal distances for wheelchair users, to enhance crowd simulations. This effort resulted in the development of a specific proxemics model, now being tested at the Pôle St Héliier Rehabilitation Center in Rennes. These evaluations will help assess its acceptability among patients with disabilities and healthcare professionals before a planned 2025 study on the effectiveness of simulator-based driving training.

4 CURRENT AND FUTURE WORK:

Until now, our experiments with the PWC simulator focused on validating it in a controlled environment. Our current work aims to assess its acceptability in an ecological context. Designed as a teaching tool for clinicians training patients acquiring a PWC, the simulator has been validated for skill transfer [7, 3]. However, we still need to determine whether clinicians see its potential use in their practice and whether patients find it valuable for learning PWC driving. To this end, we recently installed the simulator at the Rehabilitation Center of Fondation Saint Héliier in Rennes, France. The setup is user-friendly, allowing clinicians to operate it independently after a 20-minutes introduction. We aim to evaluate its usability through a demonstration scene where users navigate in a simulated crowd. Using questionnaires designed by clinicians for both their peers and patients, we seek to assess its perceived utility and gather feedback to improve the simulator. Simultaneously, the Interdisciplinary Research Center for Rehabilitation and Social Integration (CIRRS) in Quebec City, Canada, which recently acquired a similar simulator, will run the questionnaire-based evaluation of the simulator's usability.

5 CONCLUSION

After an accident or illness, some individuals face mobility challenges and need to learn to operate devices such as power wheelchairs to regain autonomy. This learning phase can be complex due to cognitive difficulties and safety concerns. Virtual reality offers a safer and more controlled environment for this process. Studies have demonstrated that greater immersion during learning and simulation significantly enhances the transfer of skills to real-world scenarios.

This principle forms the foundation of the design of the PWC simulator, which combines a mechanical platform with a HMD to deliver a highly immersive experience. To accommodate various disabilities, the platform features a seat and electronic controls

coming from commercial power wheelchairs. Designed for clinical settings, the simulator is user-friendly, enabling any clinician to operate it with minimal training.

Our current research focuses on evaluating the extent to which clinicians and patients envision using the simulator in their respective practices and learning processes. Additionally, we aim to explore aspects of social navigation. To this end, the demonstration environment includes a scene populated with a simulated crowd, allowing users to navigate freely and experience the feedback provided by the simulator.

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